

Electrical Engineering Line-up

ASI Job #: 8688
Qty. (1)
Customer: Performance Seats, Spartanburg, SC
Package Release: 05/18/26
Hardware Due: 06/08/26
Panel Complete: 06/19/26
Runoff Date: 08/18/14
Fixture Description: Blackbird 1st Row FSB Torque Lockout Station
PLC Enclosure: 24" tall x 30" wide x 8" deep ~ Electro-matic Pt. #EMRA3WM8017549A015
HMI Enclosure: 20" tall x 20" wide x 8" deep ~ Rittal Pt. #1050250 (no CNC modifications)
Support Arm Kit: N/A (Tekno swivel as described)
PLC: Allen-Bradley Compact Logics – Part #1769-L16ER-BB1B
HMI: Maple Systems 15" Smart Touchscreen – Part #CMT2158XV2
PLC IP: 172.24.0.41
HMI IP: 172.24.0.42
TC #1 IP: 172.24.0.43
Camera IP: 172.24.0.44
Subnet Mask: 255.255.0.0
Gateway: 172.24.24.1
Cycle Start Style:



Reference Job: 8470 (base hardware, cycle start)

Fixture Outline:

Automated Solutions will design and manufacture (x1) 1st Row FSB Assembly Station for the Ford Blackbird Mustang Program. The station will feature tooling with part holding, part presence sensing, pneumatic clamping & vision. The tooling will also require 180° rotate & 90° tilt (A-Side Up). The station will also include a torque reaction arm system for bolt fastening.

Machine Home Condition:

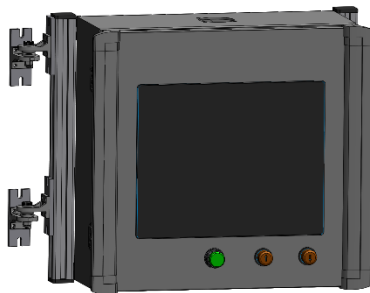
1. Clamp cylinder(s) retracted
2. Rotate home
3. Part present sensor(s) OFF

Anticipated Sequence of Operations:

- Operator loads part into fixture “B” side towards him & part present sensor verifies load.
- Operator uses Opto-touch buttons to clamp.
- Operator slides LH & RH brackets over studs.
- Operator inserts recliner rod thru recliner mechanisms on both sides.
- Operator installs LH & RH clock springs.
- Operator torque (x2) nuts on LH side & (x2) nuts on RH side
 - (x4 total) nuts with DC tool on reaction arm
 - Sequence required
 - 1st Left Front, 2nd Left Rear
 - 3rd Right Front, 4th Right Rear
 - Torque value: 35.0±5.0Nm
- Operator uses Opto-touch buttons to trigger vision check to verify recliner shaft & springs present.
 - OK allows for cycle complete (both sides pass)
 - NOK locks out the fixture until a supervisor comes to remove the part.
- Operator uses Opto-touch buttons to unclamp and remove completed assembly.

Major components on machine:

- 120VAC main power with 24VDC control voltage – include standard rotary disconnect
- (x1) Overhead LED light required ~ Banner Pt. #WLB32ZC285MQMB
 - Tie to timer relay but also include button on HMI
- (x1) Metal Power Strip ~ Amazon Pt. #B0B54N4LVQX
- (x1) Allen-Bradley Compact Logics PLC ~ Pt. #1769-L16ER-BB1B
- (x1) HMI Assembly includes all below
 - (x1) Maple Systems 15” Smart Touchscreen ~ Part #CMT2158XV2
 - (x1) Enclosure ~ Rittal Pt. #1060250 (includes sub-plate)
 - (x4) 1x1 Extrusion ~ Tekno Pt. #E0100201 x 19.50” Lg. (MBOM)
 - (x4) 1x1 End Cap ~ Tekno Pt. #H0400221 (MBOM)
 - (x2) Swivel Adapter ~ Tekno Pt. #B070051100 (MBOM)
 - (x4) End Plate ~ Tekno Pt. #B0300121 (MBOM)



- (x1) Allen-Bradley Green System Start Button ~ Pt. #800T-QT24G
 - Mounted inside HMI box (see image above)
- (x1) Allen-Bradley Auto/Manual Key Switch ~ Pt. #800T-H42A
 - Mounted inside HMI box (see image above)
- (x1) Allen-Bradley Vision Failure Bypass Key Switch ~ Pt. #800T-H42A
 - Mounted inside HMI box (see image above)
- (x1) Dual Ethernet programming port ~ Mencom Part # MEN-DP1-2RJ45-16LS
 - Mount on side of enclosure
- (x1) Stack Light ~ Banner Pt. #TL50GYRBAQ
 - (x1) Cable for stack light ~ IFM Pt. #EVC002
- (x1) E-stop Button ~ Banner Part #SSA-EB1PLYR-12ECQ8
 - (x1) IFM QD Cable for E-stop ~ Part #E11807
- (x1) E-Stop Relay ~ Banner Part #ES-FA-9AA
- (x2) Banner Opto Touch Button ~ Part #STBVR81Q6 (64192)
 - (x2) IFM QD Cable for opto touch ~ Part #EVC071

- (x1) Cycle Start Relay ~ Banner Part #AT-FM-10K (60698)
- (x1) Vision Sensor ~ Keyence Pt. #VS-L500CX
 - (x1) Power I/O Cable ~ Keyence Pt. #OP-88811
 - (x1) Ethernet Cable ~ Keyence Pt. #OP-88836
 - (x1) LED Light ~ Keyence Pt. #CA-DEW10X
 - (x1) Light Filter ~ Keyence Pt. #OP-88807
 - (x1) Power Supply ~ Keyence Pt. #CA-U4
 - (x1) Mounting Plate ~ Swivellink Pt. #AFSB-554-K-UVF
 - (x2) Standard Ball Base ~ Swivellink Pt. AFSB-1
 - (x3) Standard Ball Knuckle ~ Swivellink Pt. AFSB-2
 - (x1) Standard Link 2" ~ Swivellink Pt. AFSB-3-2
 - (x1) Standard Link 4" ~ Swivellink Pt. AFSB-3-4
- (x1) Atlas Copco Torque Tool System – Torque Controller #1 (ASI to Supply)
 - (x1) Power Focus 8 Torque controller ~ Pt. #8436280002 (put on EBOM)
 - (x1) IAM Storage Unit ~ Pt. #8436095009 (put on EBOM)
 - (x1) 90° ETV-STR61-50-10 gun ~ Pt. #8436616050 (put on EBOM)
 - (x1) 5M Gun Cable ~ Pt. #4220420405 (put on EBOM)
 - Standard bad torque & re-try key switch logic not required
 - 1st attempt is primary / 2nd, 3rd & 4th attempts are re-hit
 - Lockout after 2 total attempts / Reset counter once good torque achieved
 - (x4) P-sets required
 - (x1) LH Bracket Front Nut
 - (x1) LH Bracket Rear Nut
 - (x1) RH Bracket Front Nut
 - (x1) RH Bracket Rear Nut
 - Torque value: 35.0±5.0Nm (all torques)
 - Torque sequence required
 - 1st – LH Front, 2nd – LH Rear, 3rd – RH Front, 4th – RH Rear
 - (x1) 90° ETV-STR61-50-10 gun ~ Pt. #8436616050 (put on EBOM)

Mechanical components for reference (on MBOM):

- (x1) Custom aluminum extrusion frame
- (x4) Linak Ergo jacks & controller
- (x1) 3/8" Air Prep Asm. standard ~ SMC Pt. #FG05024
- (x1) Manifold Sub-Assembly required includes all below
 - (x1) 4-Station manifold ~ SMC Pt. #VV5QC41-04N7TFD0
 - (x1) 3 position closed valve. # VQC4301-51 **VALVE #1**
 - (1) Clamp Cylinder – Station #1
 - (x3) Blank plate # VVQ4000-10A-1 **VALVES #2, #3 & #4**
- (x1) Part present sensor for backframe
- (x2) Part present sensor for rotate home & 180°
- (x2) Position sensor for reaction arm position on X-Axis
- (x1) I/O-Link linear transducer for reaction arm position on Y-Axis
- (x2) I/O-Link Master block ~ Balluff Pt. #BNI006A
- (x1) Splitter ~ IFM Pt. #EBC113

Inputs:

- (1) Control Power On
- (1) Auto Mode Enable Key Switch
- (1) Camera bypass Key Switch
- (1) Cycle Start Relay
- (5) Part Present Sensors
- (1) I/O-Link Transducer
- (2) Adv/Ret Clamp Cylinder Switch

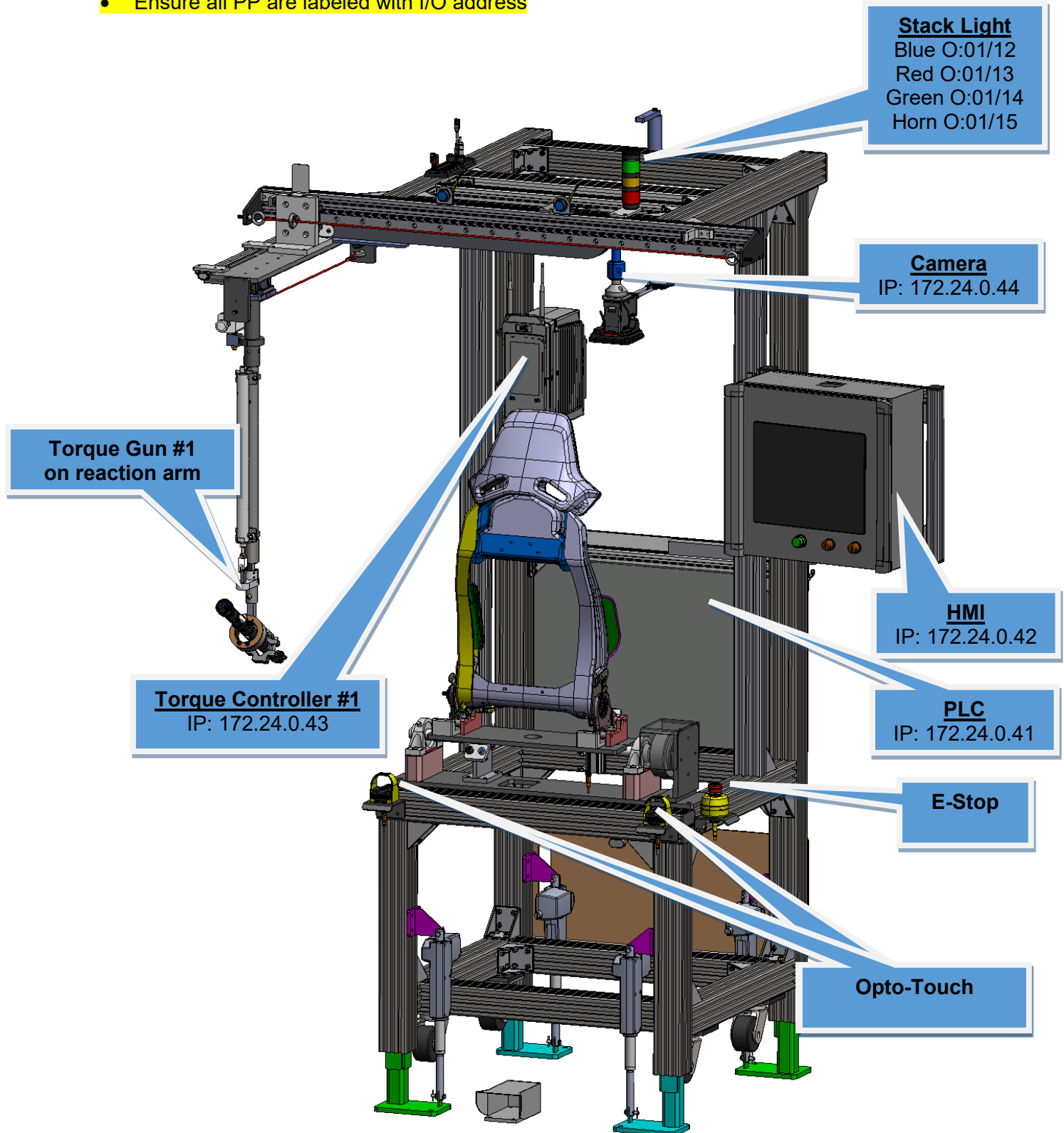
12 Inputs

Outputs:

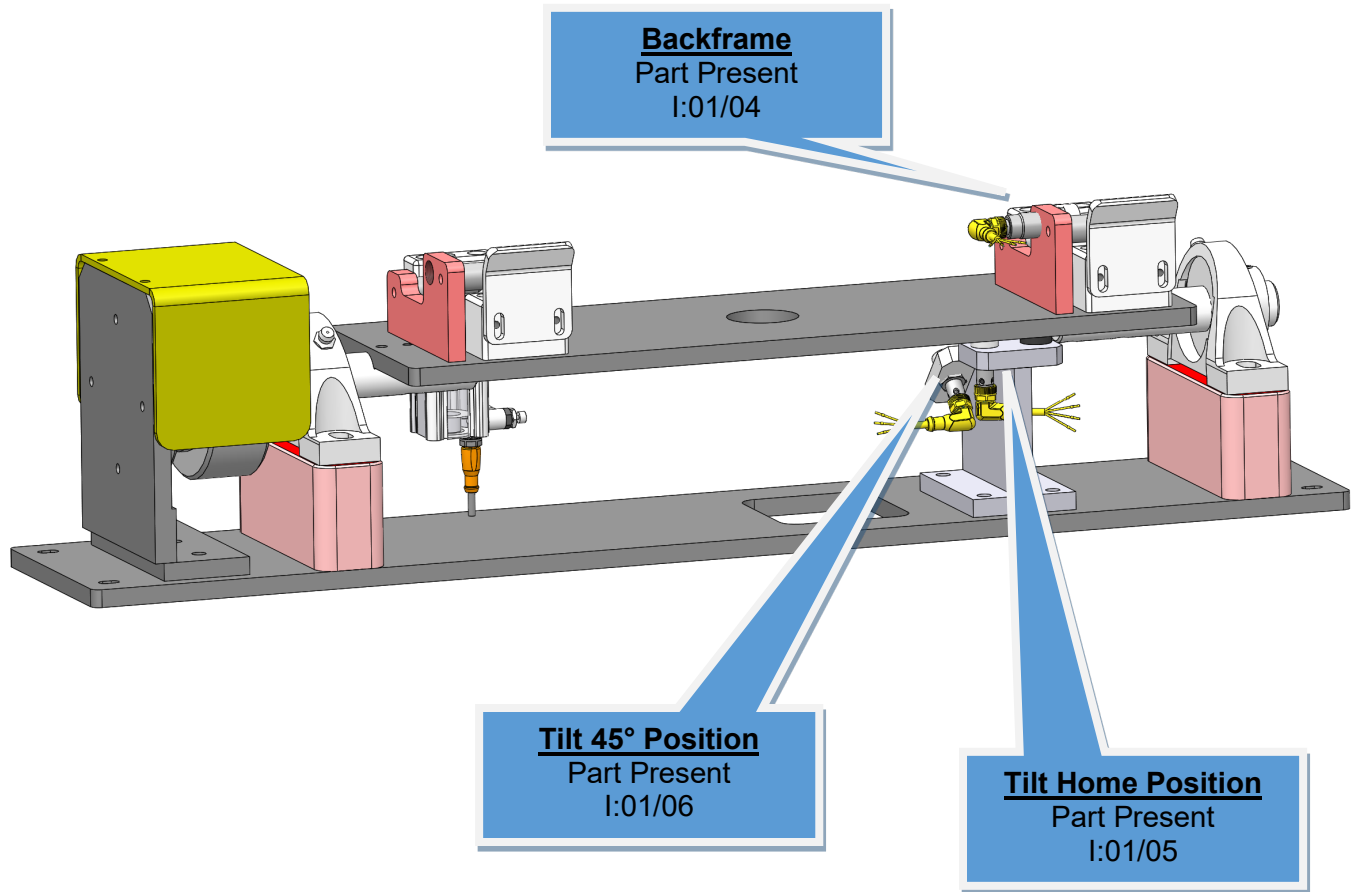
- (1) Control Power On Button Light
- (2) Adv/Ret Clamp Cylinder
- (6) Adv/Ret Spare Valves
- (4) Stack Light
- (1) Overhead Light Relay

Station Shown with Notes/Comments:

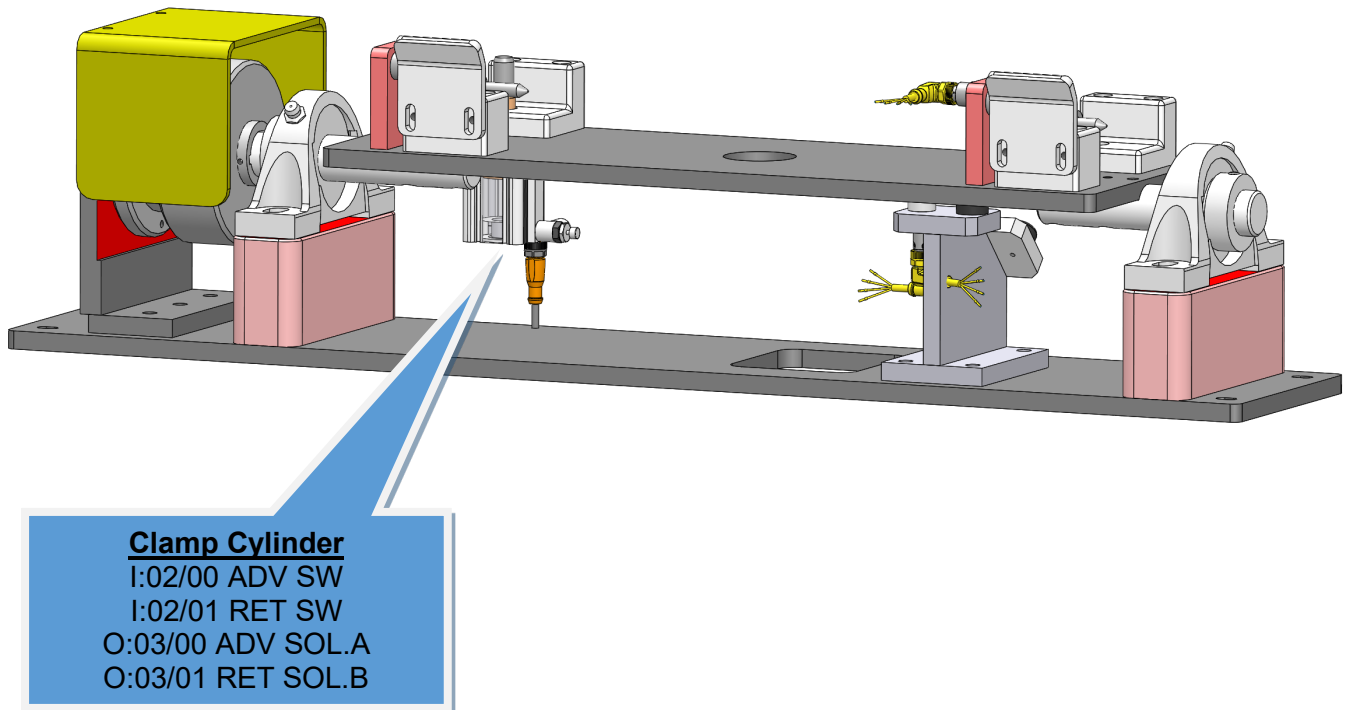
- ASI standard hardware and controls practices
- LED light board not required
- Please ensure 30% spare I/O
- Leave valves energized in their work and home states
- Lockout stickers will be required P1, E1
- Ensure all PP are labeled with I/O address



Tooling Shown with Notes/Comments:



Cylinder Fault Screen:



Torque Screen:

LH Bracket T/A Position

Part Present

BK:01/01

RH Bracket T/A Position

Part Present

BK:01/02

Fore/Aft I/O-Link Transducer

BK:01/00

LH Front Nut

RH Front Nut

LH Rear Nut

RH Rear Nut

